

Execution Environment

Author: u64127570
File: /home/u64127570/sasuser.v94/THE2.sas
SAS Platform: Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host: ODAWS01-USW2-2.ODA.SAS.COM
SAS Version: 9.04.01M7P08062020
SAS Locale: en_US
Submission Time: 4/17/2025, 10:38:58 PM
Browser Host: C-67-165-187-91.HSD1.IL.COMCAST.NET
User Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0
Application Server: ODAMID00-USW2-2.ODA.SAS.COM

Code: THE2.sas

```
/* Take Home Exam - Q2 */
```

```

1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69      /* Take Home Exam - Q2 */
70
71      data chips;
72          infile "/home/u64127570/sasuser.v94/SoggyChips.csv" dsd missover firstobs=2;
73          input time moisture;
74      run;

```

NOTE: The infile "/home/u64127570/sasuser.v94/SoggyChips.csv" is:
 Filename=/home/u64127570/sasuser.v94/SoggyChips.csv,
 Owner Name=u64127570,Group Name=oda,
 Access Permission=-rw-r--r--,
 Last Modified=17Apr2025:10:12:02,
 File Size (bytes)=231

NOTE: 24 records were read from the infile "/home/u64127570/sasuser.v94/SoggyChips.csv".
 The minimum record length was 7.
 The maximum record length was 7.

NOTE: The data set WORK.CHIPS has 24 observations and 2 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.01 seconds
memory	757.31k
OS Memory	24484.00k
Timestamp	04/18/2025 03:38:57 AM
Step Count	45
Page Faults	0
Page Reclaims	88
Page Swaps	0
Voluntary Context Switches	19
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	264

```

75
76      /* proc print data =chips; */
77      /* run; */
78
79      proc sgplot data = chips;
80          title "Chips Moisture Content vs Frying Time";
81          scatter x=time y=moisture /markerattrs=(symbol=trianglefilled color=ORANGE); *so they look like doritos :) ;
82          xaxis label= "Fry Time";
83          yaxis label= "Moisture Content (%)";
84      run;

```

NOTE: PROCEDURE SGPLOT used (Total process time):

real time	0.10 seconds
user cpu time	0.03 seconds
system cpu time	0.00 seconds
memory	8899.90k
OS Memory	30508.00k
Timestamp	04/18/2025 03:38:57 AM
Step Count	46
Page Faults	0
Page Reclaims	1732
Page Swaps	0
Voluntary Context Switches	149
Involuntary Context Switches	2
Block Input Operations	0
Block Output Operations	472

NOTE: There were 24 observations read from the data set WORK.CHIPS.

```

85      /* Looks like a reciprocal shape, as fry time increases, moisture content quickly decreases before leveling out.*/
86
87      proc corr data=chips pearson spearman;
88          var moisture time;
89      run;

```

NOTE: PROCEDURE CORR used (Total process time):

real time	0.02 seconds
user cpu time	0.03 seconds
system cpu time	0.00 seconds
memory	1787.68k
OS Memory	29860.00k
Timestamp	04/18/2025 03:38:57 AM
Step Count	47
Page Faults	0
Page Reclaims	96
Page Swaps	0
Voluntary Context Switches	3
Involuntary Context Switches	1

```
Block Input Operations      0
Block Output Operations     0
```

```
90      /* Spearman is better since the shape is not linear, my choice is supported by the test stats showing very strong
91      ! relationship for spearman vs pearson which shows moderate to strong.
92      spearman coef= -0.99302, p <0.001 */
93      data trans_chips;
94      set chips;
95      rec_moist = 1/moisture;
96      run;
```

NOTE: There were 24 observations read from the data set WORK.CHIPS.

NOTE: The data set WORK.TRANS_CHIPS has 24 observations and 3 variables.

NOTE: DATA statement used (Total process time):

```
real time      0.00 seconds
user cpu time   0.00 seconds
system cpu time 0.00 seconds
memory         942.12k
OS Memory      30376.00k
Timestamp      04/18/2025 03:38:57 AM
Step Count     48  Switch Count  2
Page Faults    0
Page Reclaims  134
Page Swaps     0
Voluntary Context Switches 15
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 264
```

```
97
98      proc reg data=trans_chips;
99          model rec_moist = time;
100          output out=reg_output p=pr_moisture;
101      run;
```

102

NOTE: The data set WORK.REG_OUTPUT has 24 observations and 4 variables.

NOTE: PROCEDURE REG used (Total process time):

```
real time      0.40 seconds
user cpu time   0.16 seconds
system cpu time 0.02 seconds
memory         11490.46k
OS Memory      39648.00k
Timestamp      04/18/2025 03:38:58 AM
Step Count     49  Switch Count  24
Page Faults    0
Page Reclaims  12752
Page Swaps     0
Voluntary Context Switches 842
Involuntary Context Switches 8
Block Input Operations 0
Block Output Operations 1176
```

```
103      data untrans_chips;
104      set reg_output;
105      pred_moist = 1/pr_moisture;
106      run;
```

NOTE: There were 24 observations read from the data set WORK.REG_OUTPUT.

NOTE: The data set WORK.UNTRANS_CHIPS has 24 observations and 5 variables.

NOTE: DATA statement used (Total process time):

```
real time      0.00 seconds
user cpu time   0.00 seconds
system cpu time 0.00 seconds
memory         945.00k
OS Memory      32168.00k
Timestamp      04/18/2025 03:38:58 AM
Step Count     50  Switch Count  2
Page Faults    0
Page Reclaims  123
Page Swaps     0
Voluntary Context Switches 17
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 264
```

```
107
108      proc sgplot data = untrans_chips;
109          title "Chips Moisture Content vs Frying Time with Fit Line";
110          scatter x=time y=moisture /markerattrs=(symbol=trianglefilled color=ORANGE); *so they look like doritos :) ;
```

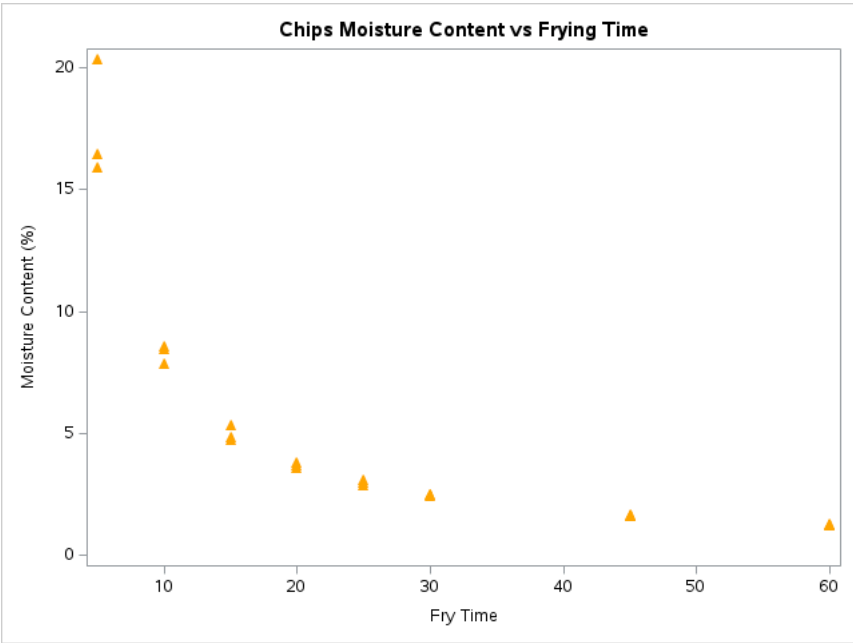
```
111      series x=time y=pred_moist;
112      xaxis label= "Fry Time";
113      yaxis label= "Moisture Content (%)";
114      run;

NOTE: PROCEDURE SGPLOT used (Total process time):
      real time          0.08 seconds
      user cpu time      0.03 seconds
      system cpu time    0.00 seconds
      memory             1545.34k
      OS Memory          32812.00k
      Timestamp          04/18/2025 03:38:58 AM
      Step Count         51  Switch Count  2
      Page Faults        0
      Page Reclaims      289
      Page Swaps         0
      Voluntary Context Switches 153
      Involuntary Context Switches 2
      Block Input Operations 0
      Block Output Operations 400
```

NOTE: There were 24 observations read from the data set WORK.UNTRANS_CHIPS.

```
115
116      /* Rsq= 0.9985 Rsadj= 0.9984
117      it was indeed reciprocal */
118
119
120      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
130
```

Results: THE2.sas



Chips Moisture Content vs Frying Time

The CORR Procedure

2 Variables: moisture time

Simple Statistics						
Variable	N	Mean	Std Dev	Median	Minimum	Maximum
moisture	24	5.33792	5.22930	3.33000	1.19000	20.33000
time	24	26.25000	17.64690	22.50000	5.00000	60.00000

Pearson Correlation Coefficients, N = 24 Prob > r under H0: Rho=0		
	moisture	time
moisture	1.00000	-0.71984 <.0001
time	-0.71984 <.0001	1.00000

Spearman Correlation Coefficients, N = 24 Prob > r under H0: Rho=0		
	moisture	time
moisture	1.00000	-0.71984 <.0001
time	-0.71984 <.0001	1.00000

Spearman Correlation Coefficients, N = 24 Prob > r under H0: Rho=0		
	moisture	time
moisture	1.00000	-0.99302 <.0001
time	-0.99302 <.0001	1.00000

Chips Moisture Content vs Frying Time

The REG Procedure
Model: MODEL1
Dependent Variable: rec_moist

Number of Observations Read	24
Number of Observations Used	24

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	1.39082	1.39082	14263.4	<.0001
Error	22	0.00215	0.00009751		
Corrected Total	23	1.39296			

Root MSE	0.00987	R-Square	0.9985
Dependent Mean	0.35496	Adj R-Sq	0.9984
Coeff Var	2.78194		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	-0.01083	0.00367	-2.95	0.0073
time	1	0.01393	0.00011668	119.43	<.0001

Chips Moisture Content vs Frying Time

The REG Procedure
Model: MODEL1
Dependent Variable: rec_moist

